

##THESIS WORK

```
list.files()
list.files("nsduh_2023")
list.files("nsduh_2023/raw", all.files = TRUE)
load("nsduh_2023/raw/NSDUH_2023.Rdata")
```

```
nsduh_raw <- data
nsduh <- nsduh_raw
rm(data)
```

```
dim(nsduh)
names(nsduh)[1:25]
library(survey)
```

####DEMOGRAPHIC VARIABLES

##RECODING AGE

```
grep("AGE", names(nsduh), value = TRUE)
table(nsduh$AGE3, useNA = "ifany")
```

```
library(dplyr)
nsduh$AGE3 <- factor(
  nsduh$AGE3,
  levels = c(1, 2, 3),
  labels = c("12-13", "14-15", "16-17")
)
table(nsduh$AGE3)
```

```
nsduh <- nsduh %>%
  filter(!is.na(AGE3))
```

##OUTCOME VARIABLE ~ MARIJUANA AND ALC USE

```
nsduh <- nsduh %>%
  mutate(
    co_use = case_when(
      CADRKMARJ %in% c(1, 5) ~ 1,
      CADRKMARJ %in% c(91, 93) ~ 0,
      CADRKMARJ %in% c(6, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
table(nsduh$CADRKMARJ, useNA = "ifany")
table(nsduh$co_use, useNA = "ifany")
```

```
nsduh$co_use <- factor(
  nsduh$co_use,
  levels = c(0, 1),
  labels = c("No", "Yes")
)
```

##SEX

```
grep("SEX", names(nsduh), value = TRUE)
nsduh <- nsduh %>%
  mutate(
    sex = case_when(
      IRSEX == 1 ~ "Male",
      IRSEX == 2 ~ "Female",
      TRUE ~ NA_character_
    )
  )
nsduh$sex <- factor(nsduh$sex, levels = c("Male", "Female"))
table(nsduh$sex, useNA = "ifany")
```

```

## RACE/ ETHNICITY
nsduh <- nsduh %>%
  mutate(
    race4 = case_when(
      NEWRACE2 == 1 ~ "Non-Hispanic White",
      NEWRACE2 == 2 ~ "Non-Hispanic Black",
      NEWRACE2 == 7 ~ "Hispanic",
      NEWRACE2 %in% c(3, 4, 5, 6) ~ "Other",
      TRUE ~ NA_character_
    )
  )

nsduh$race4 <- factor(
  nsduh$race4,
  levels = c(
    "Non-Hispanic White",
    "Non-Hispanic Black",
    "Hispanic",
    "Other"
  )
)

table(nsduh$race4, useNA = "ifany")

##POVERTY LEVEL
grep("POV", names(nsduh), value = TRUE)
nsduh <- nsduh %>%
  mutate(
    poverty = case_when(
      POVERTY3 == 1 ~ "Below poverty threshold",
      POVERTY3 == 2 ~ "Below 200% poverty threshold",
      POVERTY3 == 3 ~ "At or above 200% poverty threshold",
      TRUE ~ NA_character_
    )
  )

nsduh$poverty <- factor(
  nsduh$poverty,
  levels = c(
    "Below poverty threshold",
    "Below 200% poverty threshold",
    "At or above 200% poverty threshold"
  ),
)

table(nsduh$poverty, useNA = "ifany")

##EDUCATION
nsduh <- nsduh %>%
  mutate(
    in_school = case_when(
      EDUSCHLGO %in% c(1, 11) ~ "Currently in school",
      EDUSCHLGO == 2 ~ "Not currently in school",
      TRUE ~ NA_character_
    )
  )

nsduh$in_school <- factor(
  nsduh$in_school,
  levels = c("Not currently in school", "Currently in school")
)

table(nsduh$in_school, useNA = "ifany")

####INDEPENDENT VARIABLES
###YOUTH EXPERIENCES

```

```

##SCHOOL ENGAGEMENT- felt about going to school
nsduh <- nsduh %>%
  mutate(
    school_like = case_when(
      YESCHFLT %in% 1:4 ~ YESCHFLT,
      YESCHFLT %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
nsduh$school_like <- factor(nsduh$school_like)
levels(nsduh$school_like) <- c(
  "Liked school a lot",
  "Kind of liked school",
  "Did not like school very much",
  "Hated school"
)
table(nsduh$school_like, useNA = "ifany")

#- schoolwork was meaningful and important
nsduh <- nsduh %>%
  mutate(
    school_work_meaningful = case_when(
      YESCHWRK %in% 1:4 ~ YESCHWRK,
      YESCHWRK %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
nsduh$school_work_meaningful <- factor(nsduh$school_work_meaningful)
levels(nsduh$school_work_meaningful) <- c("Always", "Sometimes", "Seldom", "Never")
table(nsduh$school_work_meaningful, useNA = "ifany")

# - importance of school for future
nsduh <- nsduh %>%
  mutate(
    school_future_importance = case_when(
      YESCHIMP %in% 1:4 ~ YESCHIMP,
      YESCHIMP %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
nsduh$school_future_importance <- factor(nsduh$school_future_importance)
levels(nsduh$school_future_importance) <- c(
  "Very important",
  "Somewhat important",
  "Somewhat unimportant",
  "Very unimportant"
)
table(nsduh$school_future_importance, useNA = "ifany")

####BEHAVIORAL/PSYCHOSOCIAL PREDICTORS
###PEER USE NORMS

#-perceived marijuana use among peers
nsduh <- nsduh %>%
  mutate(
    peer_mj_use = case_when(
      YESTSMJ %in% 1:4 ~ YESTSMJ,
      YESTSMJ %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
nsduh$peer_mj_use <- factor(
  nsduh$peer_mj_use,
  levels = 1:4,

```

```

    labels = c(
      "None of them",
      "A few of them",
      "Most of them",
      "All of them"
    )
  )
)

table(nsduh$peer_mj_use, useNA = "ifany")

#binary for modeling
nsduh <- nsduh %>%
  mutate(
    peer_mj_use_bin = case_when(
      peer_mj_use %in% c("None of them", "A few of them") ~ "Low",
      peer_mj_use %in% c("Most of them", "All of them") ~ "High",
      TRUE ~ NA_character_
    )
  )

nsduh$peer_mj_use_bin <- factor(
  nsduh$peer_mj_use_bin,
  levels = c("Low", "High")
)

table(nsduh$peer_mj_use_bin, useNA = "ifany")
table(nsduh$peer_mj_use_bin, nsduh$co_use)

#-perceived alcohol use among peers
nsduh <- nsduh %>%
  mutate(
    peer_alc_use = case_when(
      YESTSALC %in% 1:4 ~ YESTSALC,
      YESTSALC %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )

nsduh$peer_alc_use <- factor(
  nsduh$peer_alc_use,
  levels = 1:4,
  labels = c(
    "None of them",
    "A few of them",
    "Most of them",
    "All of them"
  )
)

table(nsduh$peer_alc_use, useNA = "ifany")

nsduh <- nsduh %>%
  mutate(
    peer_alc_use_bin = case_when(
      peer_alc_use %in% c("None of them", "A few of them") ~ "Low",
      peer_alc_use %in% c("Most of them", "All of them") ~ "High",
      TRUE ~ NA_character_
    )
  )

nsduh$peer_alc_use_bin <- factor(
  nsduh$peer_alc_use_bin,
  levels = c("Low", "High")
)

table(nsduh$peer_alc_use_bin, useNA = "ifany")
table(nsduh$peer_alc_use_bin, nsduh$co_use)

```

```

#- perceived frequency of peers getting drunk at least weekly
nsduh <- nsduh %>%
  mutate(
    peer_drunkness = case_when(
      YESTSDNK %in% 1:4 ~ YESTSDNK,
      YESTSDNK %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
nsduh$peer_drunkness <- factor(
  nsduh$peer_drunkness,
  levels = 1:4,
  labels = c(
    "None of them",
    "A few of them",
    "Most of them",
    "All of them"
  )
)
table(nsduh$peer_drunkness, useNA = "ifany")

nsduh <- nsduh %>%
  mutate(
    peer_drunkness_bin = case_when(
      peer_drunkness %in% c("None of them", "A few of them") ~ "Low",
      peer_drunkness %in% c("Most of them", "All of them") ~ "High",
      TRUE ~ NA_character_
    )
  )
nsduh$peer_drunkness_bin <- factor(
  nsduh$peer_drunkness_bin,
  levels = c("Low", "High")
)
table(nsduh$peer_drunkness_bin, useNA = "ifany")
table(nsduh$peer_drunkness_bin, nsduh$co_use)

#- source of alcohol
nsduh <- nsduh %>%
  mutate(
    alcohol_source = case_when(
      CAFREWHO %in% c(1, 11) ~ "Parent/guardian",
      CAFREWHO %in% c(2, 3, 13) ~ "Other adult (21+)",
      CAFREWHO %in% c(4, 14) ~ "Underage source",
      CAFREWHO %in% c(5, 15) ~ "Took from home",
      CAFREWHO == 6 ~ "Took from someone else's home",
      CAFREWHO == 7 ~ "Other source",
      CAFREWHO %in% c(85, 91, 93, 94, 97, 98, 99) ~ NA_character_,
      TRUE ~ NA_character_
    )
  )
nsduh$alcohol_source <- factor(
  nsduh$alcohol_source,
  levels = c(
    "Parent/guardian",
    "Other adult (21+)",
    "Underage source",
    "Took from home",
    "Took from someone else's home",
    "Other source"
  )
)
table(nsduh$alcohol_source, useNA = "ifany")

```

####INDIVIDUAL LEVEL ATTITUDES

#perceived friend norms- marijuana

```
nsduh <- nsduh %>%
  mutate(
    friend_mj_disapproval = case_when(
      YEGMJMO %in% 1:3 ~ YEGMJMO,
      YEGMJMO %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
nsduh$friend_mj_disapproval <- factor(
  nsduh$friend_mj_disapproval,
  levels = 1:3,
  labels = c(
    "Neither approve nor disapprove",
    "Somewhat disapprove",
    "Strongly disapprove"
  )
)
table(nsduh$friend_mj_disapproval, useNA = "ifany")
```

#perceived friend norms- alcohol

```
nsduh <- nsduh %>%
  mutate(
    friend_alc_disapproval = case_when(
      YEGALDLY %in% 1:3 ~ YEGALDLY,
      YEGALDLY %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
nsduh$friend_alc_disapproval <- factor(
  nsduh$friend_alc_disapproval,
  levels = 1:3,
  labels = c(
    "Neither approve nor disapprove",
    "Somewhat disapprove",
    "Strongly disapprove"
  )
)
table(nsduh$friend_alc_disapproval, useNA = "ifany")
```

#perceived close friends- marijuana

```
nsduh <- nsduh %>%
  mutate(
    closefriends_mj_disapproval = case_when(
      YEFMJMO %in% 1:3 ~ YEFMJMO,
      YEFMJMO %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
nsduh$closefriends_mj_disapproval <- factor(
  nsduh$closefriends_mj_disapproval,
  levels = 1:3,
  labels = c(
    "Neither approve nor disapprove",
    "Somewhat disapprove",
    "Strongly disapprove"
  )
)
table(nsduh$closefriends_mj_disapproval, useNA = "ifany")
```

#perceived close friends - alcohol

```
nsduh <- nsduh %>%
```

```

mutate(
  closefriends_alc_disapproval = case_when(
    YEFALDLY %in% 1:3 ~ YEFALDLY,
    YEFALDLY %in% c(85, 94, 97, 98, 99) ~ NA_real_,
    TRUE ~ NA_real_
  )
)
nsduh$closefriends_alc_disapproval <- factor(
  nsduh$closefriends_alc_disapproval,
  levels = 1:3,
  labels = c(
    "Neither approve nor disapprove",
    "Somewhat disapprove",
    "Strongly disapprove"
  )
)
table(nsduh$closefriends_alc_disapproval, useNA = "ifany")

```

####MENTAL HEALTH

```

#suicidal ideation
nsduh <- nsduh %>%
  mutate(
    suicidal_ideation = case_when(
      YUSUITHK == 1 ~ 1,
      YUSUITHK == 2 ~ 0,
      YUSUITHK %in% c(3, 4, 85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
nsduh$suicidal_ideation <- factor(
  nsduh$suicidal_ideation,
  levels = c(0, 1),
  labels = c("No", "Yes")
)
table(nsduh$suicidal_ideation, useNA = "ifany")

```

#depressive symptoms

```

nsduh <- nsduh %>%
  mutate(
    depressive_symptoms = case_when(
      YODPREV == 1 ~ 1,
      YODPREV == 2 ~ 0,
      YODPREV %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
nsduh$depressive_symptoms <- factor(
  nsduh$depressive_symptoms,
  levels = c(0, 1),
  labels = c("No", "Yes")
)
table(nsduh$depressive_symptoms, useNA = "ifany")

```

####COVARIATES- PARENTAL ENVIRONMENT

```

#parental engagement
nsduh <- nsduh %>%
  mutate(
    parent_talked_drugs = case_when(
      YEPRTDNG == 1 ~ 1,
      YEPRTDNG == 2 ~ 0,
      YEPRTDNG %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )

```

```

    )
  )
  nsduh$parent_talked_drugs <- factor(
    nsduh$parent_talked_drugs,
    levels = c(0, 1),
    labels = c("No", "Yes")
  )
  table(nsduh$parent_talked_drugs, useNA = "ifany")

####PERCEIVED PARENTAL NORMS
#parental norms- marijuana
nsduh <- nsduh %>%
  mutate(
    parent_mj_disapproval = case_when(
      YEPMJMO %in% 1:3 ~ YEPMJMO,
      YEPMJMO %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
  nsduh$parent_mj_disapproval <- factor(
    nsduh$parent_mj_disapproval,
    levels = 1:3,
    labels = c(
      "Neither approve nor disapprove",
      "Somewhat disapprove",
      "Strongly disapprove"
    )
  )
  table(nsduh$parent_mj_disapproval, useNA = "ifany")

#parental norms- alcohol
nsduh <- nsduh %>%
  mutate(
    parent_alc_disapproval = case_when(
      YEPALDLY %in% 1:3 ~ YEPALDLY,
      YEPALDLY %in% c(85, 94, 97, 98, 99) ~ NA_real_,
      TRUE ~ NA_real_
    )
  )
  nsduh$parent_alc_disapproval <- factor(
    nsduh$parent_alc_disapproval,
    levels = 1:3,
    labels = c(
      "Neither approve nor disapprove",
      "Somewhat disapprove",
      "Strongly disapprove"
    )
  )
  table(nsduh$parent_alc_disapproval, useNA = "ifany")

####SURVEY DESIGN OBJECT
nsduh_design <- svydesign(
  id = ~VEREP,
  strata = ~VESTR_C,
  weights = ~ANALWT2_C,
  data = nsduh,
  nest = TRUE
)
summary(nsduh_design)

##OUTCOME + DEMOGRAPHIC
svymean(~co_use, nsduh_design, na.rm = TRUE)
prop.table(svytable(~co_use, nsduh_design)) * 100

```

```
prop.table(svytable(~sex, nsduh_design)) * 100
prop.table(svytable(~AGE3, nsduh_design)) * 100
prop.table(svytable(~race4, nsduh_design)) * 100
prop.table(svytable(~poverty, nsduh_design)) * 100
prop.table(svytable(~in_school, nsduh_design)) * 100
```

##BEHAVIORAL AND PSYCHOSOCIAL PREDICTORS

```
prop.table(svytable(~school_like, nsduh_design)) * 100
prop.table(svytable(~school_work_meaningful, nsduh_design)) * 100
prop.table(svytable(~school_future_importance, nsduh_design)) * 100
```

```
prop.table(svytable(~peer_mj_use_bin, nsduh_design)) * 100
prop.table(svytable(~peer_alc_use_bin, nsduh_design)) * 100
prop.table(svytable(~peer_drunkenness_bin, nsduh_design)) * 100
prop.table(svytable(~alcohol_source, nsduh_design)) * 100
```

##INDIVIDUAL LEVEL ATTITUDES

```
prop.table(svytable(~friend_mj_disapproval, nsduh_design)) * 100
prop.table(svytable(~friend_alc_disapproval, nsduh_design)) * 100
prop.table(svytable(~closefriends_mj_disapproval, nsduh_design)) * 100
prop.table(svytable(~closefriends_alc_disapproval, nsduh_design)) * 100
```

##MENTAL HEALTH

```
prop.table(svytable(~suicidal_ideation, nsduh_design)) * 100
prop.table(svytable(~depressive_symptoms, nsduh_design)) * 100
```

##PARENTAL ENGAGEMENT

```
prop.table(svytable(~parent_talked_drugs, nsduh_design)) * 100
prop.table(svytable(~parent_mj_disapproval, nsduh_design)) * 100
prop.table(svytable(~parent_alc_disapproval, nsduh_design)) * 100
```

```
predictors <- c(
  "AGE3", "sex", "race4", "poverty", "in_school", "school_like",
  "school_work_meaningful", "school_future_importance", "peer_mj_use_bin",
  "peer_alc_use_bin", "peer_drunkenness_bin", "friend_mj_disapproval",
  "friend_alc_disapproval", "closefriends_mj_disapproval",
  "closefriends_alc_disapproval", "suicidal_ideation", "depressive_symptoms",
  "parent_talked_drugs", "parent_mj_disapproval", "parent_alc_disapproval"
)
```

```
age_biv <- svyglm(co_use ~ AGE3, design = nsduh_design, family = quasibinomial())
sex_biv <- svyglm(co_use ~ sex, design = nsduh_design, family = quasibinomial())
race_biv <- svyglm(co_use ~ race4, design = nsduh_design, family = quasibinomial())
poverty_biv <- svyglm(co_use ~ poverty, design = nsduh_design, family = quasibinomial())
in_school_biv <- svyglm(co_use ~ in_school, design = nsduh_design, family =
quasibinomial())
school_like_biv <- svyglm(co_use ~ school_like, design = nsduh_design, family =
quasibinomial())
school_work_meaningful_biv <- svyglm(co_use ~ school_work_meaningful, design =
nsduh_design, family = quasibinomial())
school_future_importance_biv <- svyglm(co_use ~ school_future_importance, design =
nsduh_design, family = quasibinomial())
peer_mj_biv <- svyglm(co_use ~ peer_mj_use_bin, design = nsduh_design, family =
quasibinomial())
peer_alc_biv <- svyglm(co_use ~ peer_alc_use_bin, design = nsduh_design, family =
quasibinomial())
peer_drunkenness_biv <- svyglm(co_use ~ peer_drunkenness_bin, design = nsduh_design,
family = quasibinomial())
friend_mj_dis_biv <- svyglm(co_use ~ friend_mj_disapproval, design = nsduh_design, family
= quasibinomial())
friend_alc_dis_biv <- svyglm(co_use ~ friend_alc_disapproval, design = nsduh_design,
family = quasibinomial())
closefriends_mj_dis_biv <- svyglm(co_use ~ closefriends_mj_disapproval, design =
nsduh_design, family = quasibinomial())
```

```

closefriends_alc_dis_biv <- svyglm(co_use ~ closefriends_alc_disapproval, design =
nsduh_design, family = quasibinomial())
suicidal_ideation_biv <- svyglm(co_use ~ suicidal_ideation, design = nsduh_design, family
= quasibinomial())
depressive_symptoms_biv <- svyglm(co_use ~ depressive_symptoms, design = nsduh_design,
family = quasibinomial())
parents_talked_drugs_biv <- svyglm(co_use ~ parent_talked_drugs, design = nsduh_design,
family = quasibinomial())
parent_mj_dis_biv <- svyglm(co_use ~ parent_mj_disapproval, design = nsduh_design, family
= quasibinomial())
parent_alc_dis_biv <- svyglm(co_use ~ parent_alc_disapproval, design = nsduh_design,
family = quasibinomial())

biv_list <- list(
  AGE3 = age_biv,
  sex = sex_biv,
  race4 = race_biv,
  poverty = poverty_biv,
  in_school = in_school_biv,
  school_like = school_like_biv,
  school_work_meaningful = school_work_meaningful_biv,
  school_future_importance = school_future_importance_biv,
  peer_mj_use = peer_mj_biv,
  peer_alc_use = peer_alc_biv,
  peer_drunkenness = peer_drunkenness_biv,
  friend_mj_disapproval = friend_mj_dis_biv,
  friend_alc_disapproval = friend_alc_dis_biv,
  closefriends_mj_disapproval = closefriends_mj_dis_biv,
  closefriends_alc_disapproval = closefriends_alc_dis_biv,
  suicidal_ideation = suicidal_ideation_biv,
  depressive_symptoms = depressive_symptoms_biv,
  parent_talked_drugs = parents_talked_drugs_biv,
  parent_mj_disapproval = parent_mj_dis_biv,
  parent_alc_disapproval = parent_alc_dis_biv
)

library(broom)
library(purrr)
library(dplyr)

biv_table <- imap_dfr(biv_list, \(model, name) {
  tidy(model, conf.int = TRUE, exponentiate = TRUE) |>
  filter(term != "(Intercept)") |>
  mutate(predictor = name)
})

biv_table <- biv_table |>
  rename(
    level = term,
    OR = estimate,
    CI_low = conf.low,
    CI_high = conf.high,
    p = p.value
  )
print(biv_table)

biv_results <- biv_table |>
  mutate(
    OR_CI = sprintf("%.2f (%.2f--%.2f)", OR, CI_low, CI_high),
    p = round(p, 3)
  ) |>
  select(predictor, level, OR_CI, p)

biv_results

```

```
###
```

```
model_final <- svyglm(
  co_use ~ AGE3 + sex + race4 + poverty + in_school + school_like +
  school_work_meaningful + school_future_importance +
  peer_mj_use_bin + peer_alc_use_bin + peer_drunkenness_bin +
  friend_mj_disapproval + friend_alc_disapproval +
  closefriends_mj_disapproval + closefriends_alc_disapproval +
  suicidal_ideation + depressive_symptoms +
  parent_talked_drugs + parent_mj_disapproval + parent_alc_disapproval,
  design = nsduh_design,
  family = quasibinomial()
)
```

```
results_table <- tidy(model_final, conf.int = TRUE) |>
  filter(term != "(Intercept)") |>
  mutate(
    OR = exp(estimate),
    CI_low = exp(conf.low),
    CI_high = exp(conf.high),
    p = round(p.value, 3),
    OR_CI = sprintf("%.2f (%.2f--%.2f)", OR, CI_low, CI_high)
  ) |>
  select(term, OR_CI, p)
```

```
results_table
View(results_table)
```

```
#####
```

```
library(broom)
library(dplyr)
library(tidyr)
```

```
tab <- tidy(model_final, conf.int = TRUE) |>
  filter(term != "(Intercept)") |>
  mutate(
    OR = exp(estimate),
    CI_low = exp(conf.low),
    CI_high = exp(conf.high),
    OR_CI = sprintf("%.2f (%.2f--%.2f)", OR, CI_low, CI_high)
  )
```

```
tab
```

```
###LOGISTIC REGRESSION
```

```
library(caret)
set.seed(123)
features <- c(
  "AGE3", "sex", "race4", "poverty", "in_school", "school_like",
  "school_work_meaningful", "school_future_importance", "peer_mj_use_bin",
  "peer_alc_use_bin",
  "peer_drunkenness_bin", "friend_mj_disapproval", "friend_alc_disapproval",
  "depressive_symptoms", "suicidal_ideation", "parent_talked_drugs",
  "parent_mj_disapproval", "parent_alc_disapproval"
)
train_index <- sample(1:nrow(nsduh), 0.7 * nrow(nsduh))
train_data <- nsduh[train_index, ]
test_data <- nsduh[-train_index, ]
```

```
train_data_clean <- na.omit(train_data[, c("co_use", features)])
test_data_clean <- na.omit(test_data[, c("co_use", features)])
```

```
logit_model <- glm(
  co_use ~ AGE3 + sex + race4 + poverty + in_school + school_like +
  school_work_meaningful + school_future_importance + peer_mj_use_bin +
```

```

    peer_alc_use_bin +
    peer_drunkenness_bin + friend_mj_disapproval + friend_alc_disapproval +
    depressive_symptoms + suicidal_ideation + parent_talked_drugs +
    parent_mj_disapproval + parent_alc_disapproval,
    data = train_data_clean, family = binomial()
)

#probabilities
logit_prob <- predict(logit_model, test_data_clean, type = "response")

library(pROC)
logit_roc <- roc(test_data_clean$co_use, logit_prob)
logit_auc <- auc(logit_roc)

#classification metrics
logit_pred <- ifelse(logit_prob > 0.05, "Yes", "No")
library(caret)
con_mat <- confusionMatrix(
  factor(logit_pred, levels = c("No", "Yes")),
  factor(test_data_clean$co_use, levels = c("No", "Yes")),
  positive = "Yes"
)

#extract metrics
logit_results <- data.frame(
  Model = "Logistic Regression",
  AUC = round(logit_auc, 3),
  Sensitivity = round(unname(con_mat$byClass["Sensitivity"]), 3),
  Specificity = round(unname(con_mat$byClass["Specificity"]), 3),
  Precision = round(unname(con_mat$byClass["Precision"]), 3),
  F1 = round(unname(con_mat$byClass["F1"]), 3)
)

logit_results

###XG BOOSTING
library(xgboost)

#creating predictor matrixes
features <- c(
  "AGE3", "sex", "race4", "poverty", "in_school", "school_like",
  "school_work_meaningful", "school_future_importance", "peer_mj_use_bin",
  "peer_alc_use_bin",
  "peer_drunkenness_bin", "friend_mj_disapproval", "friend_alc_disapproval",
  "depressive_symptoms", "suicidal_ideation", "parent_talked_drugs",
  "parent_mj_disapproval", "parent_alc_disapproval"
)

#drop NAs from outcome and features
train_data_clean <- na.omit(train_data[, c("co_use", features)])
test_data_clean <- na.omit(test_data[, c("co_use", features)])

#build feature matrices
X_train <- model.matrix(~ . - 1, data = train_data_clean[, features])
X_test <- model.matrix(~ . - 1, data = test_data_clean[, features])

#outcome vectors
y_train_xgb <- ifelse(train_data_clean$co_use == "Yes", 1, 0)
y_test_xgb <- ifelse(test_data_clean$co_use == "Yes", 1, 0)
y_test <- test_data_clean$co_use

dtrain <- xgb.DMatrix(data = X_train, label = y_train_xgb)
dtest <- xgb.DMatrix(data = X_test, label = y_test_xgb)

```

```

#set xg parameters
params <- list(
  objective = "binary:logistic",
  eval_metric = "auc",
  max_depth = 4,
  eta = 0.05,
  subsample = 0.8,
  colsample_bytree = 0.8
)

#train xg model
xgb_model <- xgb.train(
  params = params,
  data = dtrain,
  nrounds = 200,
  watchlist = list(train = dtrain),
  verbose = 0
)

#predict probabilities
xgb_pred <- predict(xgb_model, dtest)

#compute auc
roc_xgb <- roc(y_test, xgb_pred)
auc_xgb <- auc(roc_xgb)
auc_xgb

threshold <- 0.05
pred_class_xgb <- factor(ifelse(xgb_pred > threshold, "Yes", "No"), levels = c("No",
"Yes"))

actual <- factor(y_test, levels = c("No", "Yes"))

#conf matrix
conf_matrix_xgb <- table(Predicted = pred_class_xgb, Actual = actual)
conf_matrix_xgb

TP <- conf_matrix_xgb["Yes","Yes"]
TN <- conf_matrix_xgb["No","No"]
FP <- conf_matrix_xgb["Yes","No"]
FN <- conf_matrix_xgb["No","Yes"]

sensitivity_xgb <- TP / (TP + FN)
specificity_xgb <- TN / (TN + FP)
precision_xgb <- TP / (TP + FP)

f1_xgb <- 2 * (precision_xgb * sensitivity_xgb) /
  (precision_xgb + sensitivity_xgb)

xgboost_results <- data.frame(
  Model = "XGBoost",
  AUC = round(auc_xgb, 3),
  Sensitivity = round(sensitivity_xgb, 3),
  Specificity = round(specificity_xgb, 3),
  Precision = round(precision_xgb, 3),
  F1 = round(f1_xgb, 3)
)
xgboost_results

final_results <- rbind(logit_results, xgboost_results)
final_results
rownames(final_results) <- NULL

dev.new(width = 5, height = 5)

```

```

plot(logit_roc, col="blue", lwd=3)
plot(roc_xgb, col="red", add=TRUE, lwd=3)
legend("bottomright",
      legend=c("Logistic Regression", "XGBoost"),
      col=c("blue", "red"),
      lwd=3)

```

```

#feature importance
library(ggplot2)
importance_matrix <- xgb.importance(
  feature_names = colnames(X_train),
  model = xgb_model
)
importance_matrix

ggplot(importance_matrix[1:10,],
      aes(x = reorder(Feature, Gain), y = Gain, fill = Gain)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(
    title = "XGBoost Feature Importance",
    x = "Predictor",
    y = "Importance (Gain)"
  ) +
  theme_minimal()

```

```

####VISUALS
library(ggplot2)

co_use_dist <- as.data.frame(prop.table(table(nsduh$co_use)))

ggplot(co_use_dist, aes(x = Var1, y = Freq * 100, fill = Var1)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("No" = "salmon", "Yes" = "skyblue")) +
  labs(
    title = "Prevalence of Marijuana and Alcohol Co-use",
    x = "Co-use Status",
    y = "Percentage (%)"
  ) +
  theme_minimal() +
  guides(fill = "none")

```

```

ggplot(nsduh, aes(x = peer_mj_use, fill = co_use)) +
  geom_bar(position = "fill") +
  labs(
    title = "Co-use by Peer Marijuana Use",
    x = "Peer Marijuana Use",
    y = "Proportion",
    fill = "Co-use"
  ) +
  theme_minimal()

```

```

#####BIVARIATE TABLE
library(dplyr)
library(tidyr)
library(survey)

library(dplyr)

```

```

library(tidyr)
library(survey)

make_table2_var <- function(var_name, design){

  f_var <- as.formula(paste0("~", var_name))
  f_joint <- as.formula(paste0("~", var_name, " + co_use"))

  # TOTAL %
  total_df <- as.data.frame(prop.table(svytable(f_var, design)) * 100)
  names(total_df) <- c("Category", "Total")

  # No/Yes %
  joint_df <- as.data.frame(prop.table(svytable(f_joint, design), margin = 1) * 100)
  names(joint_df) <- c("Category", "co_use", "Percent")

  joint_wide <- joint_df %>%
    pivot_wider(names_from = co_use, values_from = Percent)

  # JOIN (now clean + safe)
  out <- left_join(total_df, joint_wide, by = "Category")

  # Add variable name
  out$Variable <- var_name

  # p-value
  model <- svyglm(
    as.formula(paste0("co_use ~ ", var_name)),
    design = design,
    family = quasibinomial()
  )

  p_val <- regTermTest(model, as.formula(paste0("~", var_name)))$p
  p_formatted <- ifelse(p_val < 0.001, "<0.001", sprintf("%.3f", p_val))

  out$p_value <- rep(p_formatted, nrow(out))

  return(out)
}

predictors <- c(
  "AGE3", "sex", "race4", "poverty", "in_school",
  "school_like", "school_work_meaningful", "school_future_importance",
  "peer_mj_use_bin", "peer_alc_use_bin", "peer_drunkenness_bin",
  "friend_mj_disapproval", "friend_alc_disapproval",
  "closefriends_mj_disapproval", "closefriends_alc_disapproval",
  "suicidal_ideation", "depressive_symptoms",
  "parent_talked_drugs", "parent_mj_disapproval", "parent_alc_disapproval"
)

table2_list <- lapply(predictors, make_table2_var, design = nsduh_design)
table2 <- bind_rows(table2_list)

table2_clean <- table2 %>%
  mutate(
    Total = round(Total, 1),
    No = round(No, 2),
    Yes = round(Yes, 2)
  ) %>%
  select(
    Variable,
    Category,
    Total,
    No,
    Yes,
    p_value
  )

```

```
)  
  
table2_clean <- table2_clean %>%  
  group_by(Variable) %>%  
  mutate(  
    Variable = ifelse(row_number() == 1, Variable, ""),  
    p_value = ifelse(row_number() == 1, p_value, "")  
  ) %>%  
  ungroup()  
  
write.csv(table2_clean, "Table2.csv", row.names = FALSE)
```